studocu

MST practice 2023

Derivative Securities (University of Melbourne)



Scan to open on Studocu

Studocu is not sponsored or endorsed by any college or university
Downloaded by Joshua Pacifici (jpac14alt@gmail.com)

FNCE 30007: Derivative Securities

Practice Mid-Semester Test, SEM 2 2023

Maximum possible score: 15 marks

Question 1 (1 mark)

On 9th September, Sheldon has closed out his short position in 6 S&P500 index futures contracts by trading with Amy. In so doing, Amy has increased her existing future positions in S&P500 index contracts by 6. What is the resultant change in the trading volume?

a) Decrease by 12
b) Increase by 6
c) Decrease by 6
d) No change
e) None of the other answers

Question 2 (1 mark)

James Harden has received the following tip from his financial advisor: "James, the price of copper will be, very likely, plummeting in the coming 4 weeks: I'd strongly recommend you to get in on it". Trusting his advisor's forecast, Harden is willing to take a 1 million dollar position using November copper futures, whose contract size is 1000 pounds. On September 7th, 2021, when Harden opens his position, the current spot price and November futures price for copper are $280 and $285 per pound, respectively and he needs to post an initial margin of $12,500 per contract. He is also informed that the maintenance margin per contract is $10,000. At the end of the first day of trading, the November futures contract settles at $287 per pound. The balance of Harden's margin account at the close on September 7th, 2021 is, therefore:

A. $440,000
B. $998,000
C. $1,160,000
D. $840,000
E. None of the other answers

Question 3 (1 mark)

On March 1 a commodity's spot price is $60 and its August futures price is $59. On July 1 the spot price is $64 and the August futures price is $63.50. A company entered into futures contracts on March 1 to hedge its purchase of the commodity on July 1. It closed out its position on July 1. What is the effective price (after taking account of the hedging) paid by the company?

a) $59.50
b) $60.50
c) $61.50
d) $63.50
e) $64.00

Downloaded by Joshua Pacifici (jpac14alt@gmail.com)

Question 4 (1 Mark)

Which of the following is NOT true:

a. If the cost of carry for an investment asset is positive, and the futures price of the asset is lower than the spot price an arbitrage opportunity is available (ignoring transactions costs).
b. If all companies in an industry do not hedge, a company is likely to reduce its risk by hedging
c. Arguments against hedging by companies point to the possibility for shareholders to hedge their own portfolios, if they so desire.
d. Reducing the beta of an equity portfolio to zero through the appropriate position in stock index futures does not guarantee the risk-free rate of return on the hedged portfolio
e. One reason for using stoic index futures to hedge an equity portfolio is to avoid the liquidation of the portfolios when it is expected to perform poorly

Question 5 (1 Mark)

Today is September 15th and you have just opened futures positions on the December corn contract to hedge a sale of corn scheduled for November 15th. On September 15th the futures price for the December contract is $8.9 while the spot price is $8.1. On November 15th the spot price of wheat has decreased to $7.5 and the futures price on the December contract settles at $6.9. On November 15th you also close out your futures position. Assuming that you would like to hedge 90% of your underlying exposure, which of the following statements is correct?

a) Your total gain/loss per unit from the hedged position is $1.2.
b) Your effective price received for each unit of corn is $8.9.
c) Your gain/loss per unit from the hedged position is $1.4 which is the same as the increase in basis.
d) More than one of other statements is correct.
e) None of the other statements is correct.

Question 6 (1 mark)

A company is naturally long iron ore. To hedge price risk the company takes a short position in copper futures contracts expiring on June 1. Which of the following statements correctly characterizes the company's position:

A. It faces no basis risk if they are certain they can sell iron ore spot on June 1
B. The futures position actually increases the risk of the spot position if the correlation between iron ore spot price and copper futures price is not perfect
C. As long as the correlation between iron ore spot price and copper futures price is =+1 the optimal hedge ratio for the company is always 1
D. It faces basis risk even if they are certain they can sell iron ore spot on June 1
E. None of the above statements is correct.

Downloaded by Joshua Pacifici (jpac14alt@gmail.com)

Question 7 (1 mark)

Peter Bol specializes in arbitraging the spot and forward stock markets. Today he is looking at one particular stock price and at forward contracts written on it. The stock is expected to pay no dividends or any other income in the foreseeable future. By Peter's calculations, the 1-year forward price should be $105 but, on his screen, he observes that the 1-year forward is quoted at $104.
Which of the following statement is correct?

A. Peter should borrow the price of the stock, buy one share of the stock and enter into a long forward contract to buy the stock in one year. In so doing, he will realize an arbitrage profit of $1 immediately.
B. Peter should short the stock, invest the proceeds of the short sale at the risk-free rate and enter into a long forward contract to buy the stock in one year. In so doing, he will realize an arbitrage profit of $1 immediately.
C. Peter should short the stock, invest the proceeds of the short sale at the risk-free rate and enter into a short forward contract to sell the stock in one year. In so doing, he will realize an arbitrage profit of $1 at the end of one-year period.
D. Peter should borrow the price of the stock, buy one share of the stock and enter into a short forward contract to sell the stock in one year. In so doing, he will realize an arbitrage profit of $1 at the end of one-year period.
E. None of other statements is correct.

Question 8 (1 mark)

Which of the following is correct?

a) The optimal hedge ratio is always 1 if there is no maturity mismatch.
b) The optimal hedge is always 1 when there is no basis risk.
c) To determine the optimal hedge ratio, one needs to use the standard deviation of futures prices and the standard deviation of the spot prices of the asset underlying the futures contract.
d) More than one of the other statements is correct.
e) None of the other statements is correct.

Question 9 (1 mark)

Bloomberg Commodity News reports that the level of copper inventories has recorded a very large and unexpected increase. All other things equal, what impact will such news have on prices?

A. The 6-month copper futures price as a percentage of the copper spot price declines
B. The copper spot price as a percentage of the 6-month copper futures price increases
C. The 6-month copper futures price as a percentage of the copper spot price stays the same
D. The copper convenience yield increases
E. Neither of the impacts described in the other statements

Downloaded by Joshua Pacifici (jpac14alt@gmail.com)

Question 10 (1 mark)

A company will buy 1000 units of a certain commodity in one year from today. It decides to hedge 80% of its exposure using futures contracts. The spot price and the futures price are currently $100 and $90, respectively. The spot price and the futures price in one year turn out to be $112 and $110, respectively. What is the average price paid for the commodity?

A. $92
B. $96
C. $102
D. $106
E. $100

Question 11 (2 marks)

Tony Stark manages an equity portfolio valued at 300,000,000 USD which has a beta of -0.7. Given his forecast for the overall market, Tony seeks to change its beta to -0.4 from September 25th, 2021 to November 25th, 2021. Tony is going to use December S&P500 index futures contracts maturing on December 25th, 2021 to achieve this objective. As the analyst assisting the fund manager, you are given the following information:

- S&P500 spot index as on September 25, 2021 = 4,520
- The dividend yield on the S&P500 index = 1.55% per annum (continuously compounded)
- The US risk-free rate = 2.65% per annum (continuously compounded)
- Each futures contract is on $250 times the index
- Correlation between the S&P500 spot returns and S&P500 futures returns = 1
- S&P500 futures return standard deviation = 1.00
- S&P500 spot return standard deviation = 1.00

What should you recommend Tony Stark to do in order to achieve his objective stated above (round number of contracts to the nearest integer)?

a) Tony should long 106 December contracts.
b) Tony should short 106 December contracts.
c) Tony should long 185 December contracts.
d) Tony should short 185 December contracts.
e) None of the other strategies is appropriate.

Question 12 (1 mark)

ABC Ltd., a Chinese company, is expecting to export equipment to Australia in December. The transaction will be settled in Australian Dollar. Which of the following futures position between Chinese Yuan and Australian Dollar is appropriate for ABC Ltd. to hedge the foreign exchange risk?

a) The company should long futures on the Chinese Yuan.
b) The company should short futures on the Chinese Yuan.
c) The company should long futures on the Australian Dollar.
d) More than one of the other strategies is appropriate.
e) None of the other strategies is appropriate.

Downloaded by Joshua Pacifici (jpac14alt@gmail.com)

Question 13 (2 marks)

At the end of the trading day on September 10th, 2021, the ASX 200 index closes at 7200 while the ASX 200 Index Futures with expiration on March 25, 2022, settles at 7060. The spot index pays a dividend yield of 3.5% per annum (continuously compounded) while the risk-free interest rate in Australia can be safely assumed to be zero at this time. Assuming that transactions costs are negligible and that the index multiplier is $1, which one of the following is a correct statement?

a. An arbitrageur can go long the futures, short the spot, invest the proceeds at the risk free rate and realize an arbitrage profit of $4.79 on September 10.
b. The futures and the spot are fairly priced and, hence, there is no arbitrage opportunity
c. An arbitrageur can go short the futures, borrow funds at the risk free rate to go long the spot and realize an arbitrage profit of $4.79 on September 10.
d. An arbitrageur can go long the futures, short the spot, invest the proceeds at the risk free rate and realize an arbitrage profit of $4.79 on March 25.
e. None of the other statements is correct.

Downloaded by Joshua Pacifici (jpac14alt@gmail.com)

THIS PAGE HAS BEEN LEFT INTENTIONALLY BLANK

SPACE FOR ROUGH WORK

Examiners will NOT take into account anything written on this page.

Downloaded by Joshua Pacifici (jpac14alt@gmail.com)

THIS PAGE HAS BEEN LEFT INTENTIONALLY BLANK

SPACE FOR ROUGH WORK

Examiners will NOT take into account anything written on this page.

Downloaded by Joshua Pacifici (jpac14alt@gmail.com)

THIS PAGE HAS BEEN LEFT INTENTIONALLY BLANK

SPACE FOR ROUGH WORK

Examiners will NOT take into account anything written on this page.

Downloaded by Joshua Pacifici (jpac14alt@gmail.com)

FORMULAE SHEET for Mid-Sem Test

Effective Price (short hedge): $P_t = S_t + F_0 - F_t$

$$b_t = S_t - F_t$$

$$h^* = \frac{\rho_{sf}\sigma_s}{\sigma_f}$$

$$N^* = \frac{h^* Q_A}{Q_F} \qquad N^* = \frac{h^* V_A}{V_F}$$

$$N^* = \beta \frac{V_A}{V_F}$$

$$N^* = \left(\beta - \beta^*\right)\frac{V_A}{V_F}$$

$$F_0 = S_0 e^{rT}$$

$$F_0 = \left(S_0 - I\right)e^{rT}$$

$$F_0 = S_0 e^{(r-q)T}$$

$$F_0 = S_0 e^{(r-r_f)T}$$

$F_0 = (S_0 + U)e^{rT}$ for investment commodities

$F_0 = S_0 e^{(r+u)T}$ for investment commodities

$F_0 \le \left(S_0 + U\right)e^{rT}$ for consumption commodities

$F_0 \le S_0 e^{(r+u)T}$ for consumption commodities

$$F_0 e^{yT} = \left(S_0 + U\right)e^{rT}$$

$$F_0 = S_0 e^{cT}$$

$F_t = S_t e^{(c-y)(T-t)}$

Downloaded by Joshua Pacifici (jpac14alt@gmail.com)